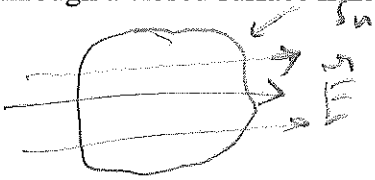
Practice Problems
Calculus-Based Physics II

Name: Key **Date:** _____

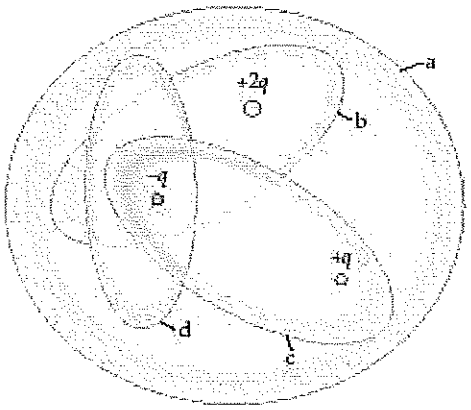
Chapter 36 Part C Electric Fields

1) If the electric flux through a closed surface is zero, the electric field strength must also be zero.

A) True
(B) False



2) The figure shows four Gaussian surfaces (a, b, c and d) surrounding a distribution of charges.



A) Which Gaussian surfaces have an electric flux of $+q/\varepsilon_0$ through them?

$\oint \vec{E} \cdot d\vec{A} = \frac{Q_{in}}{\varepsilon_0} = \frac{(+2q) + (-q)}{\varepsilon_0} = \frac{+q}{\varepsilon_0}$

(b)

B) Which Gaussian surfaces have no electric flux through them?

$\oint \vec{E} \cdot d\vec{A} = \frac{Q_{in}}{\varepsilon_0} = \frac{(+q + (-q))}{\varepsilon_0} = 0$

(c)

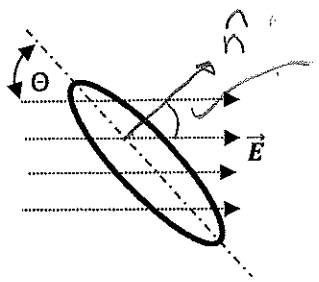
C) Determine the electric flux through the Gaussian surface "a".

$\oint \vec{E} \cdot d\vec{A} = \frac{Q_{in}}{\varepsilon_0} = \frac{(+2q + (-q) + 5)}{\varepsilon_0} = \frac{+2q}{\varepsilon_0}$

3) An uncharged conductor has a hollow cavity inside of it. Within this cavity there is a positive charge of 10 μC that does not touch the conductor. There are no other charges in the vicinity. Which statement about this conductor is true? (There may be more than one correct choice.)

~~A)~~ The inner surface of the conductor carries a charge of -10 μC and its outer surface carries no excess charge.
~~B)~~ The inner and outer surfaces of the conductor each contain charges of -5 μC.
~~C)~~ The net electric field within the material of the conductor points away from the +10 μC charge.
(D) The outer surface of the conductor contains +10 μC of charge and the inner surface contains -10 μC.
~~E)~~ Both surfaces of the conductor carry no excess charge because the conductor is uncharged.

4) A circular hoop (r = 6 cm) is in a uniform electric field with a magnitude of 400 N/C, the plane of the hoop makes an angle of 45° to the electric field lines. Determine the electric flux through the hoop.



$\phi = 90^\circ - \theta = 45^\circ$

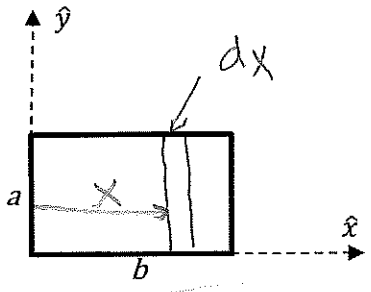
$A = \pi r^2$

$$\Phi_E = \oint \vec{E} \cdot d\vec{A} = |\vec{E}| A \cos\phi = |\vec{E}| (\pi r^2) \cos(\phi)$$

$$\Phi_E = 3.2 \ N m^2/C$$

5) A rectangular shape is in the x-y plane as shown in the figure below, If the electric field strength has a x-axis dependency and points along the z-axis, determine the flux through the rectangle.

$\vec{E}(x) = (mx^2)\hat{z}$



$$\Phi_E = \int \vec{E} \cdot d\vec{A} = \int |\vec{E}| \underbrace{\hat{z} \cdot \hat{z}}_{1} dA$$

$$\Phi_E = \int_0^b (m x^2)(a)\, dx$$

$d\vec{A} = (a\,dx)\,\hat{z}$

$\hat{n}$ - direction

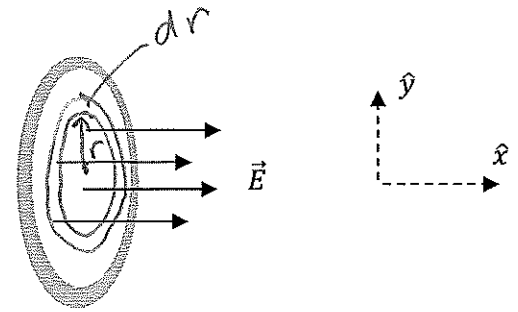
$$\Phi_E = ma \int_0^b x^2 dx$$

$$\Phi_E = \frac{ma}{3}\left[x^3 \right]_0^b$$

$$\Phi_E = \frac{mab^3}{3}$$

6) A circular hoop with radius R is in a non-uniform electric field that is perpendicular to the hoop's area, determine the electric flux going through the hoop given the changing electric field below.

$$\vec{E}(r) = (a + b\sqrt{r})\hat{x}$$



$$d\vec{A} = [(2\pi r)\,dr]\,\hat{x}$$

$$\Phi_E = \int \vec{E} \cdot d\vec{A}$$

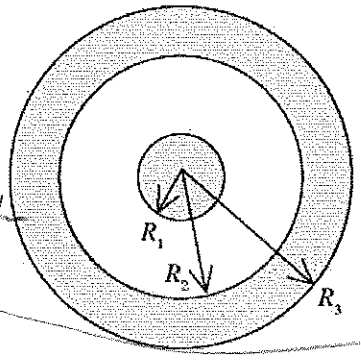
$$\Phi_E = \int_0^R (a + b\sqrt{r})(2\pi r)\,\underbrace{\hat{x}\cdot\hat{x}}_{1}\,dr$$

$$\Phi_E = 2\pi \int_0^R (ar + br^{3/2})\,dr$$

$$= 2\pi \left[\frac{1}{2}r^2 + \frac{2b}{5}r^{5/2}\right]_0^R$$

$$\Phi_E = 2\pi \left[\frac{aR^2}{2} + \frac{2b\,r^{5/2}}{5}\right)$$

7) Two concentric spheres are shown in the figure. The inner sphere is a solid conductor and carries a charge of 5.0 μC positive, uniformly distributed over its outer surface. The outer sphere is a conducting shell, a negative charge of 8.0 μC is placed on the outer shell. No other charges are present. The radii shown in the figure have the values $R_1 = 10.0$ cm, $R_2 = 20.0$ cm, and $R_3 = 30.0$ cm.



A) Find the total excess charge on the inner and outer surfaces of the conducting sphere.

Inner → $Q_{inside} = 5\mu C$ negative

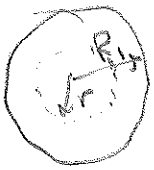
Outer → $Q_{out} = (-8\mu C) + (5\mu C) = 3\mu C$ negative

Find the magnitude and direction of the electric field at the following distances r from the center of the inner sphere:

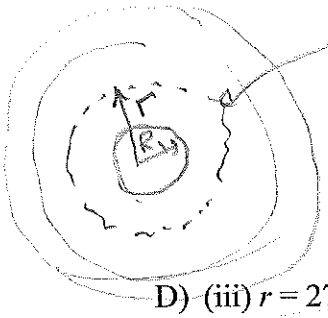
B) (i) $r = 9.5$ cm,

$\oint \vec{E} \cdot d\vec{A} = \frac{Q_{in}}{\varepsilon_0}$ $\quad Q_{in} = 0\,C$ inside conductor

$E = \frac{Q_{in}}{A\varepsilon_0}$ $\quad E = 0\,N/C$



C) (ii) $r = 15.0$ cm,



$r = 15cm = 0.15m$

$Q_{in} = 5\mu C$

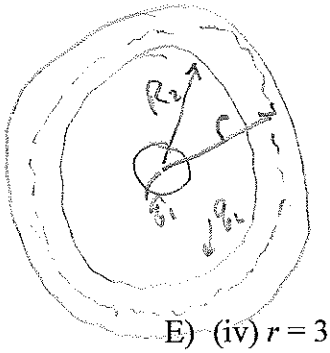
$A = 4\pi r^2$

$\oint \vec{E} \cdot d\vec{A} = \frac{Q_{in}}{\varepsilon_0}$

$|\vec{E}| = \frac{Q_{in}}{\varepsilon_0 A} = \frac{5 \times 10^{-6} C}{\varepsilon_0 (4\pi (0.15m)^2}$

$|\vec{E}| = 2 \times 10^6\,N/C$

D) (iii) $r = 27.0$ cm,



$Q_{in} = (-5\mu C) + (+5\mu C) = 0$

$|\vec{E}| = \frac{Q_{in}}{\varepsilon_0 A} = 0\,N/C$

E) (iv) $r = 35.0$ cm.

$A = 4\pi r^2$

$Q_{in} = 3\mu C$ neg

$\oint \vec{E} \cdot d\vec{A} = \frac{Q_{in}}{\varepsilon_0}$

$|\vec{E}| = \frac{Q_{in}}{A\varepsilon_0} = \frac{3 \times 10^{-6} C}{4\pi (0.35m)^2 \varepsilon_0} = 2.2 \times 10^5\,N/C$

8) A spherical, non-conducting object with radius R = 20 cm carries a total charge Q = 15 μC distributed uniformly throughout the volume of the sphere. What is the magnitude of the electric field at a distance *r* = 12 cm from the center of the sphere?

need Volume Charge density

$$\rho = \frac{Q_{Total}}{V} = \frac{15\times10^{-6}\,C}{4/3\,\pi R^3}$$

$$\rho = 4.48\times10^{-4}\,C/m^3$$

$$\oint \vec{E}\,d\vec{A} = \frac{Q_{in}}{\varepsilon_0}$$

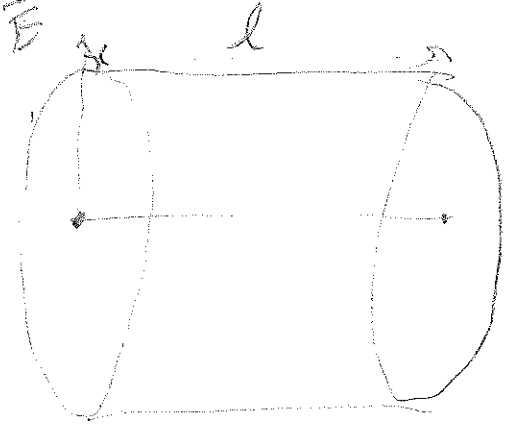
$$Q_{in} = \rho V = \rho(4/3\,\pi r^3)$$

$$A = 4\pi r^2$$

$$|\vec{E}| = \frac{Q_{in}}{\varepsilon_0 A} = \frac{4/3\,\pi r^3 \rho}{\varepsilon_0(4\pi r^2)} = \frac{\rho r}{3\varepsilon_0}$$

$$|\vec{E}| = 2.0\times10^4\,N/C$$

9) Determine the magnitude of the electric field a distance of r = 20 cm above a long wire with a total length of 50 m and a total charge of 4 x 10^{-6} C using Gauss's Law.



$\vec{E}$ is perpendicular to surface of cylinder

$$A = 2\pi r \ell$$

$\vec{E}$-field is parallel to ends of cylinder

$$\Phi_{Ends} = 0.$$

$$\lambda = \frac{Q_{Total}}{\ell} = 8\times10^{-8}\,C/m$$

$$Q_{in} = \lambda \ell$$

$$\oint \vec{E}\cdot d\vec{A} = \frac{Q_{in}}{\varepsilon_0}$$

$$|\vec{E}| = \frac{Q_{in}}{\varepsilon_0 A} = \frac{\lambda \ell}{\varepsilon_0 2\pi r \ell} = \frac{\lambda}{2\pi\varepsilon_0 r} = 7.2\times10^3\,N/C$$

Answers

1) B
2) A) b B) c C) $2q/\varepsilon_o$
3) D
4) 3.2 Nm2/C
5) $\Phi = mab^3/3$
6) See solutions
7) A) inter surface Q = 5 μC negative outer surface Q = 3 μC negative
   B) $\Phi = 0$ E = 0 C) 2×10^6 N/C D) E = 0 E) 2.2×10^5 N/C
8) 2.0×10^6 N/C
9) 7.2×10^3 N/C